Preliminaries

Benjamin J. Clark

Fall 2025

1 Special common operations

1.1 Modulo arithmetic

When working in finite groups (will be defined later) we want the operation of addition to remain closed. To accomplish this, we make addition "wrap back around" on itself. This is called modulo arithmetic.

Theorem 1.1 (The quotient remainder theorem)**.** *Given any integer n and positive integer d, there exists unique integers q and r with $0 \leq r < d$ such that*

$$n = dq + r.$$

Definition 1.2. Given an integer n and a positive integer d, if $n = dq + r$ for q r integers such that $0 \leq r < d$ then

$$n \text{ div } d = q$$
$$n \bmod d = r.$$

Example 1.3. Time is a common place to see modulo arithmetic. When we talk about hours in a day, we are doing modulo arithmetic. Consider it is 8pm, and we want to know what hour it will be 100 hours from now. Instead of directly counting, we can take 99 mod 24 = 3, then add the 3 hours to 8pm (which is really 20 hours into the day) to get 11pm. So hours in a day is modulo 24.

We can use the quotient remainder theorem to take proofs involving an integer and break them into a finite number of cases based on the divisor d.

Definition 1.4. For any real number x, the **absolute value of** x, denoted $|x|$, is defined as follows:

$$|x| = \begin{cases} x & \text{if } x \geq 0 \\ -x & \text{if } x < 0. \end{cases}$$

With this definition, we have the following common properties for all real numbers x, y:

1. $-|x| \leq x \leq |x|$.
2. $|-x| = |x|$.
3. $|x+y| \leq |x| + |y|$.

1.2 Complex numbers

Real numbers have a shortcoming when solving equations, consider $x^2+1=0$. What are the roots of x? Solving for x gives $\pm\sqrt{-1}$, but we know the square root function can't take negative numbers. So this equation has no solutions in the real numbers.

To fix this, we will define a new number system called the complex numbers.

Definition 1.5. The complex numbers are defined as a pair of real numbers a, b in the form $a+bi$ where $i=\sqrt{-1}$. Symbolically

$$\mathbb{C} = \{a+bi \mid a, b \in \mathbb{R} \text{ and } i = \sqrt{-1}\}.$$

For complex numbers $a+bi$ and $c+di$ we have

$$\begin{aligned}(a+bi)+(c+di) &= (a+c)+(b+d)i \\ (a+bi)(c+di) &= (ac-bd)+(ad+bd)i.\end{aligned}$$

So, addition is done by adding across and multiplication is done by expanding the two binomials, keeping in mind that $i^2=-1$.

Complex numbers form an algebraically complete field, which for now just means all polynomials of degree n have n roots.

2 Proofs

2.1 Direct proof and counterexample.

Definition 2.1. An integer n is **even** if, and only if, n equals twice some integer. An integer n is **odd** if, and only if, n equals twice some integer plus 1. Symbolically this gives:

$$\begin{aligned}n \text{ is even} &\iff n = 2k \text{ for some integer } k. \\ n \text{ is odd} &\iff n = 2k+1 \text{ for some integer } k.\end{aligned}$$

Definition 2.2. An integer n is **prime** if, and only if, $n > 1$ and for all positive integers r and s, if $n = rs$, then either r or s equals n. An integer n is **composite** if, and only if, $n > 1$ and $n = rs$ for some integers r and s with $1 < r < n$ and $1 < s < n$.

Note that these two definitions give us ways to break up the integers. An integer can only be even or odd, but not both. Similarly, a positive integer can be either prime or composite, but not both.

Procedure 2.3 (Proving existential and disproving universal statements)**.** If you need to prove an existential statement of the form,

$$\exists x \in D \text{ such that } Q(x)$$

then all it takes is finding one $x \in D$ that makes $Q(x)$ true. Similarly, if you need to disprove a universal statement of the form

$$\forall x \in D, P(x),$$

then that is the same as proving the negation which is

$$\exists x \in D \text{ such that } \sim P(x).$$

So to disprove the universal statement we need to find one $x \in D$ such that $P(x)$ is false. When dealing with integers I would recommend $-1, 0, 1$ as easy options to start with.

Procedure 2.4 (Proving universal or disproving existential statements)**.** If we want to prove a statement of the form,

$$\forall x \in D, \text{ if } P(x) \text{ then } Q(x).$$

then below are a couple of the main type of proof strategies that can be used.

1. **Method of exhaustion**: If the domain D is finite or if you can split it into a finite number of cases, then the method of exhaustion can be used by checking each case.

   For example, when dealing with integers using that they are either even or odd is a common strategy.

2. **Direct proof**: Here we start with our if $P(x)$ as an assumption and use definitions, previous theorems, and other results to directly get to our conclusion. This is the most standard proof method and is often used with other strategies.

   The general form of this proof will start with

   > Let $x \in D$ such that $P(x)$ is true, then ...

   To show that a result is true for all elements in a set we need to use a variable to represent an arbitrary element and work only off the properties that all elements in the set have.

3. **Proof by contradiction**: Another common proof strategy is proof by contradiction. Here we are going to assume $\sim Q(x)$ and $P(x)$ then try to reach a logical contradiction. This strategy can be helpful since we get the extra information on x that $\sim Q(x)$ is true.

Theorem 2.5. *The sum of any two even integers is even.*

Proof. Suppose $m, n \in \mathbb{Z}$ such that m, n are even, then by the definition of being even $m = 2r$ and $n = 2s$ for some integers $r, s \in \mathbb{Z}$. With this we have

$$m + n = 2r + 2s = 2(r + s).$$

Let $t = r + s$ and note that t is an integer since it is the sum of integers. Hence, $m + n = 2t$ which is the definition of being even. $\square$

While a lot of problems involving concepts from even/odd and primes seem easy to prove, they can be deceptively hard. A well known open problem in math is as follows

Conjecture 2.6 (Goldbach conjecture)**.** *Let $n \in \mathbb{Z}$ such that $n > 2$, then n is the sum of two prime numbers.*

2.2 General tips

Copy the statement of the theorem to be proved on your paper

This makes the theorem statement available for reference to anyone reading the proof.

Clearly mark the beginning of your proof with the word Proof

This word separates general discussion about the theorem from its actual proof.

Make your proof self-contained

Explain the meaning of each variable used in your proof. Begin proofs by introducing the initial variables to be used. This is similar to declaring variables and their data types at the beginning of a computer program.

Common words to start a proof are Assume, Let, If, Suppose.

Write your proof in complete, grammatically correct sentences

This does not mean that you should avoid using symbols and shorthand abbreviations, just that you should incorporate them into sentences.

Don't start your sentences with symbols. Try and prep the symbols with short phrases of where the sentence is going. Common words are Following, Since, This gives, We have, Now.

Keep your reader informed about the status of each statement in your proof

Your reader should never be in doubt about whether something in your proof has been assumed or established, or is still to be deduced. If something is assumed, preface it with a word like Suppose or Assume. If it is still to be shown, preface it with words like, We must show that or In other words, we must show that. This is especially important if you introduce a variable in rephrasing what you need to show. (See Common Mistakes.)

Give a reason for each assertion in your proof

Each assertion in a proof should come directly from the hypothesis of the theorem, or follow from the definition of one of the terms in the theorem, or be a result obtained earlier in the proof, or be a mathematical result that has previously been established or is agreed to be assumed. Indicate the reason for each step of your proof using phrases such as by hypothesis, by definition of ... by theorem ... and so forth.

It is best to refer to definitions and theorems by name or number. If you need to state one in the body of your proof, avoid using a variable when you write it because otherwise your proof could end up with a variable that has two conflicting meanings.

Proofs in more advanced mathematical contexts often omit reasons for some steps because it is assumed that students either understand them or can easily figure them out for themselves. However, in a course that introduces mathematical proof, you should make sure to provide the details of your arguments because you cannot guarantee that your readers have the necessary mathematical knowledge and sophistication to supply them on their own.

Include the "little words and phrases" that make the logic of your arguments clear

When writing a mathematical argument, especially a proof, indicate how each sentence is related to the previous one. Does it follow from the previous sentence or from a combination of the previous sentence and earlier ones? If so, start the sentence with the word Because or Since and state the reason why it follows, or write Then, or Thus, or So, or Hence, or Therefore, or Consequently, or It follows that, and include the reason at the end of the sentence.

If a sentence expresses a new thought or fact that does not follow as an immediate consequence of the preceding statement but is needed for a later part of a proof, introduce it by writing Observe that, or Note that, or Recall that, or But, or Now.

Sometimes in a proof, it is desirable to define a new variable in terms of previous variables. In such a case, introduce the new variable with the word Let.

Display equations and inequalities

The convention is to display equations and inequalities on separate lines to increase readability, both for other people and for ourselves so that we can more easily check our work for accuracy.

2.3 Common Mistakes

Arguing from examples

Looking at examples is one of the most helpful practices a problem solver can engage in, and is encouraged by all good mathematics teachers. However, it is a mistake to think that a general statement can be proved by showing it to be true for some individual cases. A property referred to in a universal statement may be true in many instances without being true in general.

Using the same letter to mean two different things

Think of the "scope" of a mathematical variable as covering the entire proof. Make sure to not double use the same letter or symbol, unless you clearly redefine it.

Jumping to a conclusion

To jump to a conclusion means to allege the truth of something without giving an adequate reason. Especially for early proofs, always justify every step you take.

Assuming what is to be proved

To assume what is to be proved is a variation of jumping to a conclusion. This can be difficult with early proof classes, as sometimes a theorem can be used on a proof which itself relies on the validity of what is trying to be proved. This can generally be avoided by justifying every step of the proof.

Use of any when the correct word is some

Sometimes the word some acts like an any in a statement, and other times it acts like a there exists. I generally only use some when writing the phrase "[statement is true] for some [value]".

Misuse of the word if

Another common error is not serious in itself, but it reflects imprecise thinking that sometimes leads to problems later in a proof. This error involves using the word if when the word because is really meant. Using the word "if" can sometimes lead to doubt on whether the value is known. I like to use "if" in proofs when I am doing a proof with cases.

2.4 Example proofs

Lemma 2.7. *If $p \in \mathbb{Z}$ is even, then p^2 is even.*

Proof. Let $p \in \mathbb{Z}$ be even, then by the definition of even $p = 2k$ for some $k \in \mathbb{Z}$. Now

$$p^2 = (2k)^2 = 4k^2 = 2(2k^2).$$

Letting $x = 2k^2$ we have $p^2 = 2x$ giving p^2 is even. $\square$

Theorem 2.8. *$\sqrt{2}$ is irrational.*

Proof. Assume for contradiction that $\sqrt{2}$ is rational, then $\sqrt{2} = \frac{p}{q}$ for $p, q \in \mathbb{Z}$ where $q \neq 0$ and $\gcd\{p, q\} = 1$. This gives

$$\sqrt{2} = \frac{p}{q} \implies 2 = \frac{p^2}{q^2} \implies 2q^2 = p^2$$

which implies p^2 is even. Because p^2 is even p must be even. Following p is even $p = 2k$ for some $k \in \mathbb{Z}$. Now

$$\begin{aligned} 2q^2 = p^2 &\implies 2q^2 = (2k)^2 \\ &\implies 2q^2 = 4k^2 \\ &\implies q^2 = 2k^2. \end{aligned}$$

Therefore q is also even. However, this is a contradiction since we assumed $\gcd\{p, q\} = 1$. Thus $\sqrt{2}$ is irrational. □

Theorem 2.9. *For every integer n, $2n - 1$ is odd.*

Here are three different ways to prove this theorem.

Proof. If $n \in \mathbb{Z}$ is odd, then by the definition of odd $n = 2k + 1$ for some $k \in \mathbb{Z}$. Now

$$2n - 1 = 2(2k + 1) - 1 = 4k + 2 - 1 = 2(2k) + 1.$$

Let $x = 2k$ which is an integer. Therefore by the definition of being odd $2n - 1 = 2x + 1$ is odd.

If $n \in \mathbb{Z}$ is even, then by the definition of even $n = 2k$ for some $k \in \mathbb{Z}$. Now

$$2n - 1 = 2(2k) - 1 = 4k - 1 = 4k - 1 + 2 - 2 = 2(2k - 1) + 1.$$

Let $x = 2k - 1$ which is an integer. Therefore by the definition of being odd $2n - 1 = 2x + 1$ is odd. □

Proof. Assume for contradiction that $2n - 1$ is even for some integer $n \in \mathbb{Z}$, then $2n - 1 = 2k$ for some $k \in \mathbb{Z}$. Now

$$2n - 1 = 2k \implies 2n = 2k + 1 \implies n = k + \frac{1}{2}.$$

However this is a contradiction since we assumed n, k to be integers but $1/2$ is not an integer. Thus $2n - 1$ is odd. □

Proof. Let $n \in \mathbb{Z}$, then

$$2n - 1 = 2n + 2 - 2 - 1 = 2(n - 1) + 1.$$

Let $x = n - 1$ which is an integer. Therefore $2n - 1 = 2x + 1$ is odd. □

Theorem 2.10. *For every integer m, if m is even, then $3m + 5$ is odd.*

Proof. Let $m \in \mathbb{Z}$ such that m is even, then $m = 2k$ for some $k \in \mathbb{Z}$. Now

$$3m + 5 = 6k + 5 = 6k + 4 + 1 = 2(3k + 2) + 1.$$

Let $x = 3k + 2$ which is an integer. Therefore $3m + 5 = 2x + 1$ is odd. $\square$

Theorem 2.11. *If k is any odd integer and m is any even integer, then $k^2 + m^2$ is odd.*

Proof. Let $k, m \in \mathbb{Z}$ such that k is odd and m is even, then $k = 2a+1$ and $m = 2b$ for some $a, b \in \mathbb{Z}$. Now

$$k^2 + m^2 = (2a + 1)^2 + (2b)^2 = 4a^2 + 4a + 1 + 4b^2 = 2(2a^2 + 2a + 2b^2) + 1.$$

Let $x = 2a^2 + 2a + 2b^2$ which is an integer. Therefore $k^2 + m^2 = 2x + 1$ is odd. $\square$

To show the statement "There exists an integer $m \geq 3$ such that $m^2 - 1$ is prime." is false we can take the negation and show it is true.

Theorem 2.12. *For all integers m, if $m \geq 3$ then $m^2 - 1$ is composite.*

Proof. Let $m \in \mathbb{Z}$ such that $m \geq 3$, then

$$m^2 - 1 = (m + 1)(m - 1).$$

Following that $m + 1$ and $m - 1$ are greater than 1 we have that $m^2 - 1$ is composite with factors of $m + 1$ and $m - 1$. $\square$

To show the statement "There exists an integer n such that $6n^2 + 27$ is prime." is false we can take the negation and show it is true.

Theorem 2.13. *For all integers n, $6n^2 + 27$ is composite.*

Proof. Let $n \in \mathbb{Z}$, then

$$6n^2 + 27 = 3(2n^2 + 9).$$

Following 3 and $2n^2+9$ are integers greater than 1 we have $6n^2+27$ is composite with factors 3 and $2n^2 + 9$. $\square$

To show the statement "There exists an integer $k \geq 4$ such that $2k^2 - 5k + 2$ is prime." is false we can take the negation and show it is true.

Theorem 2.14. *For all integers $k \geq 4$, $2k^2 - 5k + 2$ is composite.*

Proof. Let $k \in \mathbb{Z}$ such that $k \geq 4$, then

$$2k^2 - 5k + 2 = (2k - 1)(k - 2).$$

Following that $2k - 1$ and $k - 2$ are integers greater than 1 for $k \geq 4$ we have $2k^2 - 5k + 2$ is composite with factors $2k - 1$ and $k - 2$. $\square$

Theorem 2.15. *Prove that the sum of any 3 consecutive integers is divisible by 3.*

Proof. Let $n \in \mathbb{Z}$, then 3 consecutive integers are $n - 1$, n, $n + 1$. Now

$$(n - 1) + n + n + 1 = 3n$$

which is divisible by 3. $\square$

2.5 Mathematical Induction

Definition 2.16 (Principal of mathematical induction)**.** Let $P(n)$ be a property that is defined for integers n, and let a be a fixed integer. Suppose the following two statements are true:

1. $P(a)$ is true.
2. For every integer $k \geq a$, if $P(k)$ is true, then $P(k+1)$ is true.

Then the statement

$$\text{for every integer } n \geq a, P(n)$$

is true.

Procedure 2.17 (Method of proof by mathematical induction)**.** To prove a statement of the form

For every integer $n \geq a$, a property $P(a)$ is true

we can use mathematical induction. To do this, first we prove the base case. Which is show $P(a)$ is true, where a is the initial value the statement should hold on.

Then we need to prove the inductive step. Assume the result holds for some $k \geq a$, then we want to show that the result holds for $k+1$.

Once you have shown these two things, the original statement is proven.

Example 2.18. Prove that for $n \geq 1$ we have

$$\sum_{k=1}^{n} k = \frac{n(n+1)}{2}.$$

Proof. For the base case, $n = 1$, we have

$$\sum_{k=1}^{1} k = 1 = \frac{1(2)}{2}.$$

Now for the inductive step we assume that the result holds for some $m \geq 1$, then

$$\begin{aligned}\sum_{k=1}^{m+1} k &= (m+1) + \sum_{k=1}^{m} k \\ &= (m+1) + \frac{m(m+1)}{2} \\ &= \frac{(m+1)(m+2)}{2}.\end{aligned}$$

□

2.6 Exercises

1. Prove that there is an integer $n > 5$ such that $2^n - 1$ is prime.
2. Prove that for every integer n, if $(n-1)/2$ is an integer, then n is odd.
3. Prove that the difference between the squares of any two consecutive integers is odd.
4. Prove the following by induction
   (a) $1^2 + 2^2 + \cdots + n^2 = \frac{n(n+1)(2n+1)}{6}$ for all $n \geq 1$.
   (b) $\sum_{k=1}^{n+1} k2^k = n2^{n+2} + 2$.
   (c) $\sum_{k=1}^{n} k(k!) = (n+1)! - 1$ for all $n \geq 1$.

2.7 Solutions

1. Prove that there is an integer $n > 5$ such that $2^n - 1$ is prime.

   Choose $n = 7$, then $2^7 - 1 = 127$ which is prime.

2. Prove that for every integer n, if $(n-1)/2$ is an integer, then n is odd.

   Proof. Let n be an integer such that $(n-1)/2$ is also an integer, then $(n-1)/2 = k$ for some integer k. Now,

$$(n-1)/2 = k \implies n - 1 = 2k \implies n = 2k + 1.$$

   Following the definition of odd numbers, n is odd. □

3. Prove that the difference between the squares of any two consecutive integers is odd.

   Proof. Let $n \in \mathbb{Z}$, then

$$(n+1)^2 - n^2 = n^2 + 2n + 1 - n^2 = 2n + 1.$$

   By definition of odd numbers $2n + 1$ is odd giving that $(n+1)^2 - n^2$ is odd. □

4. Prove the following by induction
   (a) $1^2 + 2^2 + \cdots + n^2 = \frac{n(n+1)(2n+1)}{6}$ for all $n \geq 1$.

Proof. Proceed by induction, for the base case of $n = 1$ we have

$$1^2 = 1 = \frac{1(2)(3)}{6}.$$

For the inductive step, assume the result holds for some $n \geq 1$, then

$$\begin{aligned} 1^2 + 2^2 + \cdots + n^2 + (n+1)^2 &= \frac{n(n+1)(2n+1)}{6} + (n+1)^2 \\ &= \frac{2n^3 + 3n^2 + n}{6} + \frac{6n^2 + 12n + 6}{6} \\ &= \frac{2n^3 + 9n^2 + 13n + 6}{6} \\ &= \frac{(n+1)(n+2)(2n+3)}{6}. \end{aligned}$$

□

(b) $\sum_{k=1}^{n+1} k2^k = n2^{n+2} + 2.$

Proof. Proceed by induction, for the base case of $n = 1$ we have

$$\sum_{k=1}^{2} k2^k = 1(2^1) + 2(2^2) = 10 = 1(2^{1+2}) + 2.$$

For the inductive step assume the result holds for some $n \geq 1$, then

$$\begin{aligned} \sum_{k=1}^{n+2} k2^k &= (n+2)2^{n+2} + \sum_{k=1}^{n+1} k2^k \\ &= (n+2)2^{n+2} + n2^{n+2} + 2 \\ &= (2n+2)2^{n+2} + 2 \\ &= (n+1)2^{n+3} + 2. \end{aligned}$$

□

(c) $\sum_{k=1}^{n} k(k!) = (n+1)! - 1$ for all $n \geq 1$.

Proof. Proceed by induction, for the base case of $n = 1$ we have

$$\sum_{k=1}^{1} k(k!) = 1(1!) = 1 = 2! - 1.$$

For the inductive step assume the result holds for some $n \geq 1$, then

$$\begin{aligned} \sum_{k=1}^{n+1} k(k!) &= (n+1)(n+1)! + \sum_{k=1}^{n} k(k!) \\ &= (n+1)(n+1)! + (n+1)! - 1 \\ &= (n+2)(n+1)! - 1 \\ &= (n+2)! - 1. \end{aligned}$$

□

3 Set theory

3.1 Definition and basic properties of sets

Definition 3.1. A set is defined as a collection of elements. These elements can be (almost) anything, including other sets. There is no implicit order to the elements of a set, and duplicates are ignored.

Definition 3.2. One way to build sets is with **set-roster notation**, which is where we list all elements of the set or list the first few once the pattern is clear to the reader. For example, $\{1, 2, 3\}$ and $\{1, 2, 3, \dots\}$.

Another way to build sets is with **set-builder notation**, which is written as $\{x \in S \mid P(x)\}$ this is read as "x in S such that $P(x)$ is true" where $P(x)$ is some property of the statement that x must satisfy. For example, $\{x \in \mathbb{R} \mid x \geq 3\}$ which is the set of all real numbers that are greater than or equal to 3. Note that instead of the $|$ it is also common to use, : this can be especially helpful in situations involving absolute values like $\{x \in \mathbb{R} : |x| < 1\}$.

The most common sets when working with numbers are $\mathbb{N}$ the natural numbers, $\mathbb{Z}$ the integers, $\mathbb{Q}$ the rational numbers, and $\mathbb{R}$ the real numbers. We will generally exclude using $\mathbb{N}$ because it has two definitions that are used about the same, which are the nonnegative integers and the positive integers. The integers are the whole numbers, including negatives. We will use $\mathbb{Z}^+$ for the positive integers and $\mathbb{Z}^{\geq 0}$ for the nonnegative integers. Another set we care about is the **empty set**, which is defined to be the set of no elements, it is denoted $\emptyset$.

When working with sets, we often care about the idea of a **subset**, which is defined as a set that contained in another set and is denoted $A \subseteq B$. If this containment is strict, that is B contains more elements then A, we write $A \subset B$ and this is called a **proper subset**.

Two sets are called equal if and only if they contain all the same elements.

Definition 3.3. Given elements a, b, the symbol (a, b) denotes the **ordered pair** made from a and b with the added information that a is first and b is second.

Two ordered pairs are equal if and only if their first and second components are the same. That is, for $(a, b) = (c, d)$ we need $a = c$ and $b = d$.

The above definition can be extended to arbitrary dimension.

Definition 3.4. Let n be a positive integer and let $x_1, \dots, x_n$ be elements. The **ordered n-tuple**, $(x_1, \dots, x_n)$, consists of the above elements with the ordering that x_1 comes before x_2 and so on.

We can apply this idea of ordered pairs to construct a multiplication like operation for sets.

Definition 3.5. Given sets $A_1, A_2, \ldots, A_n$ the Cartesian product of $A_1, \ldots, A_n$ denoted $A_1 \times A_2 \times \ldots \times A_n$, is the set of all ordered tuples $(a_1, \ldots, a_n)$ where $a_1 \in A_1$ and so on. Symbolically, we can write this as

$$A_1 \times A_2 \times \ldots \times A_n = \{(a_1, a_2, \ldots, a_n) | a_1 \in A_1, a_2 \in A_2, \ldots, a_n \in A_n\}.$$

Definition 3.6 (Common set operations)**.** Let A and B be subsets of a universal set U.

1. The **union** of A and B, denoted $A \cup B$, is the set of all elements that are in at least one of A or B.

2. The **intersection** of A and B, denoted $A \cap B$, is the set of all elements that are common to both A and B.

3. The **difference** of B minus A, denoted $B - A$, is the set of all elements that are in B and not in A.

4. The **complement** of A, denoted A^c, is the set of all elements in U that are not in A.

Definition 3.7 (Unions and intersections of an indexed collection of sets)**.** Given sets $A_0, A_1, A_2, \ldots$ that are subsets of a universal set U and given a nonnegative integer n,

$$\bigcup_{i=0}^{n} A_i = \{x \in U | x \in A_i \text{ for some } i = 0, 1, \ldots, n\}$$
$$\bigcap_{i=0}^{n} A_i = \{x \in U | x \in A_i \text{ for every } i = 0, 1, \ldots, n\}.$$

Definition 3.8 (Interval notation)**.** Given real numbers a and b with $a \leq b$:

$$(a, b) = \{x \in \mathbb{R} | a < x < b\} \qquad [a, b] = \{x \in \mathbb{R} | a \leq x \leq b\}$$
$$(a, b] = \{x \in \mathbb{R} | a < x \leq b\} \qquad [a, b) = \{x \in \mathbb{R} | a \leq x < b\}.$$

Definition 3.9. Two sets are called **disjoint** if and only if they have no elements in common.

We can extend this definition to a list of sets $A_1, A_2, A_3, \ldots$ and say that all the A_i sets are **mutually disjoint** if all pairs of the sets are disjoint.

Definition 3.10. A collection of nonempty sets $\{A_1, A_2, \ldots\}$ is a **partition** of a set A if and only if

1. A is the union of all the A_i

2. The sets $A_1, A_2, \ldots$ are mutually disjoint.

Definition 3.11. Given a set A, the **power set** of A, denoted $\mathcal{P}(A)$, is the set of all subsets of A.

Theorem 3.12 (Set inclusion principals)**.** *Let A, B, C be sets, then*

1. $A \cap B \subseteq A$
2. $A \subseteq A \cup B$
3. *If* $A \subseteq B$ *and* $B \subseteq C$*, then* $A \subseteq C$.

Theorem 3.13 (Set Identities)**.** *Let $A, B, C \subset U$ where U is the universal set, then*

1. *Commutative laws:*
$$A \cup B = B \cup A \text{ and } A \cap B = B \cap A.$$
2. *Associative laws:*
$$(A \cup B) \cup C = A \cup (B \cup C) \text{ and } (A \cap B) \cap C = A \cap (B \cap C).$$
3. *Distributive laws:*
$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$$
and
$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C).$$
4. *Identity laws:*
$$A \cup \emptyset = A \text{ and } A \cap U = A.$$
5. *Complement laws:*
$$A \cup A^c = U \text{ and } A \cap A^c = \emptyset.$$
6. *Double complement law:*
$$(A^c)^c = A.$$
7. *Idempotent laws:*
$$A \cup A = A \text{ and } A \cap A = A.$$
8. *Universal bound laws:*
$$A \cup U = U \text{ and } A \cap \emptyset = \emptyset.$$
9. *De Morgan's laws:*
$$(A \cup B)^c = A^c \cap B^c \text{ and } (A \cap B)^c = A^c \cup B^c.$$
10. *Absorption laws:*
$$A \cup (A \cap B) = A \text{ and } A \cap (A \cup B) = A.$$
11. *Complements of U and $\emptyset$:*
$$U^c = \emptyset \text{ and } \emptyset^c = U.$$
12. *Set difference law:*
$$A - B = A \cap B^c.$$

Theorem 3.14. *Let A be a set, then $\emptyset \subseteq A$.*

Theorem 3.15. *For all sets A, B, C, if $A \subseteq B$ and $B \subseteq C^c$, then $A \cap C = \emptyset$.*

3.2 Proofs on sets

Procedure 3.16. Given sets, A, B to prove that, $A \subseteq B$ we need to show for any element in A that element is in B. The standard way to do this is

1. Suppose that a is an unknown, but particular element of A.
2. Show that a is an element of B.

Procedure 3.17. Given sets A, B the standard way to prove that, $A = B$ is to prove $A \subseteq B$ and that $B \subseteq A$.

One way to think of set relations is as a for all statement. That is for all sets we have this relation. So, to disprove a set relation we need to construct a counterexample.

Example 3.18. Disprove the following. For all sets A, B, C we have

$$(A - B) \cup (B - C) = A - C.$$

To start building the counterexample think about what each side is saying. On the right we have A with everything from C removed. However on the left we have A with everything from B removed or B with everything from C removed. So if we have an element in C that is in A but not in B then it will break our equality. For example

$$\begin{aligned} A &= \{1\} \\ B &= \{2\} \\ C &= \{1\} \\ (A - B) \cup (B - C) &= \{1, 2\} \\ (A - C) &= \emptyset. \end{aligned}$$

Theorem 3.19. *Let $n \in \mathbb{Z}$ such that $n \geq 0$, if X is a set with n elements, then $\mathcal{P}(X)$ has 2^n elements.*

Proof. Let $n = 0$, then X is the set with no elements so its only subset is itself giving $\mathcal{P}(X)$ has 1 element. Now assume the result holds for some $m \geq 0$. If X has $m+1$ elements, then it has at least one element $z \in X$. Consider $X - \{z\}$ this is a set with m elements so by our induction hypothesis $\mathcal{P}(X - \{z\})$ has 2^m subsets. Now to build $\mathcal{P}(X)$ we can either add z to each subset in $\mathcal{P}(X - \{z\})$ or not. This gives two possible choices for each element. Thus $\mathcal{P}(X)$ has $2(2^m) = 2^{m+1}$ elements. $\square$

3.3 Exercises

1. Let $\mathbb{R}$ be the universal set and let

$$\begin{aligned} A &= \{x \in \mathbb{R} \mid -3 \leq x \leq 0\}, \\ B &= \{x \in \mathbb{R} \mid -1 < x < 2\}, \\ C &= \{x \in \mathbb{R} \mid 6 < x \leq 8\}. \end{aligned}$$

   Find the following

(a) $A \cup B$

(b) B^c

(c) $A^c \cap B^c$

(d) $(A \cap B)^c$

2. Let

$$A = \{x \in \mathbb{Z} | x = 10b - 3 \text{ for some integer } b\}$$
$$B = \{z \in \mathbb{Z} | z = 18c + 16 \text{ for some integer } c\}.$$

Prove or give a counter example to $A \subset B$, $B \subset A$, and $A = B$.

3. Is $\{\{a, d, e\}, \{b, c\}, \{d, f\}\}$ a partition of $\{a, b, c, d, e, f\}$. If so, justify. If not how can you make it one?

4. Let $A = \{1, 2\}$ and $B = \{2, 3\}$, then find the following.

(a) $\mathcal{P}(\emptyset)$.

(b) $\mathcal{P}(A)$.

(c) $\mathcal{P}(A \cap B)$.

(d) $\mathcal{P}(A \cup B)$.

5. Prove the following

$$(A - B) \cup (C - B) = (A \cup C) - B$$

6. Find a counterexample to the following

(a) $(A \cup B) \cap C = A \cup (B \cap C)$

(b) If $B \cup C \subseteq A$, then $(A - B) \cap (A - C) = \emptyset$.

3.4 Solutions

1. Let $\mathbb{R}$ be the universal set and let

$$A = \{x \in \mathbb{R} \mid -3 \leq x \leq 0\},$$
$$B = \{x \in \mathbb{R} \mid -1 < x < 2\},$$
$$C = \{x \in \mathbb{R} \mid 6 < x \leq 8\}.$$

Find the following

(a) $A \cup B$ $[-3, 2)$ or $\{x \in \mathbb{R} \mid -3 \leq x < 2\}$.

(b) B^c $(-\infty, -1] \cup [2, \infty)$ or $\{x \in \mathbb{R} \mid x \leq -1 \text{ or } x \geq 2\}$.

(c) $A^c \cap B^c$ $(-\infty, -3] \cup [2, \infty)$

(d) $(A \cap B)^c$ $(-\infty, -1] \cup (0, \infty)$

2. Let

$$A = \{x \in \mathbb{Z} | x = 10b - 3 \text{ for some integer } b\}$$
$$B = \{z \in \mathbb{Z} | z = 18c + 16 \text{ for some integer } c\}.$$

Prove or give a counter example to $A \subseteq B$, $B \subseteq A$, and $A = B$.

We know $-3 \in A$ since $-3 = 10(0) - 3$. For -3 to be in B we need $-3 = 18c + 16 \implies -19 = 18c$ for some integer $c \in \mathbb{Z}$. However, this is a contradiction since -19 is not divisible by 18. This gives that A is not a subset of B. Note this also shows that A is not equal to B.

Taking a similar approach to showing that B is not a subset of A. Let $16 \in B$, then for 16 to be in A we need $16 = 10b - 3 \implies 19 = 10b$. However, this can't happen since 19 is not divisible by 10.

3. Is $\{\{a, d, e\}, \{b, c\}, \{d, f\}\}$ a partition of $\{a, b, c, d, e, f\}$. If so, justify. If not, how can you make it one?

This is not a partition. While the sets $\{a, d, e\}, \{b, c\}, \{d, f\}$ do union to $\{a, b, c, d, e, f\}$, we have a duplicate d in sets $\{d, f\}$ and $\{a, d, e\}$.

To fix this, we can simply remove the duplicate d. An example would be $\{\{a, d, e\}, \{b, c\}, \{f\}\}$.

4. Let $A = \{1, 2\}$ and $B = \{2, 3\}$, then find the following.

(a) $\mathcal{P}(\emptyset)$
Note that the power set is a set containing sets and that it always contains the set itself. In this case, that is the only subset giving

$$\mathcal{P}(\emptyset) = \{\emptyset\}.$$

(b) $\mathcal{P}(A)$.

$$\mathcal{P}(A) = \{\{1, 2\}, \{1\}, \{2\}, \emptyset\}.$$

(c) $\mathcal{P}(A \cap B)$.

$$\mathcal{P}(A \cap B) = \{\{2\}, \emptyset\}.$$

(d) $\mathcal{P}(A \cup B)$.

$$\mathcal{P}(A) = \{\{1, 2, 3\}, \{1, 2\}, \{1, 3\}, \{2, 3\}, \{1\}, \{2\}, \{3\}, \emptyset\}.$$

5. Prove the following $(A-B)\cup(C-B)=(A\cup C)-B$

   Proof.
   $$\begin{aligned}(A-B)\cup(C-B) &= (A\cap B^c)\cup(C\cap B^c) && \text{(Set difference law)}\\ &= (A\cup C)\cap B^c && \text{(Distributive law)}\\ &= (A\cup C)-B. && \text{(Set difference law)}\end{aligned}$$
   □

6. Find a counterexample to the following
   (a) $(A\cup B)\cap C = A\cup(B\cap C)$
   $$A=\{1\},\ B=\{2\},\ C=\{3\}$$
   $$(A\cup B)\cap C=\emptyset$$
   $$A\cup(B\cap C)=\{1\}.$$
   (b) If $B\cup C\subseteq A$, then $(A-B)\cap(A-C)=\emptyset$.
   $$A=\{1\},\ B=\emptyset,\ C=\emptyset$$
   $$B\cup C=\emptyset\subseteq A$$
   $$(A-B)\cap(A-C)=\{1\}.$$

4 Relations

4.1 Properties of relations

Definition 4.1. Let A and B be sets A relation R from A to B is a subset of $A\times B$. We use xRy to mean $(x,y)\in R$. The set A is called the domain and B is called the co-domain.

Definition 4.2. Let R be a relation from A to B. Define the inverse relation R^{-1} from B to A as
$$R^{-1}=\{(y,x)\in B\times A \mid (x,y)\in R\}.$$

Example 4.3. Let $A=\{2,3,4\}$ and $B=\{2,6,8\}$, then
$$A\times B=\{(2,2),(2,6),(2,8),(3,2),(3,6),(3,8),(4,2),(4,6),(4,8)\}.$$
If we have
$$R=\{(2,2),(2,6),(2,8),(3,6),(4,8)\},$$
then
$$R^{-1}=\{(2,2),(6,2),(8,2),(6,3),(8,4)\}.$$

Definition 4.4. A relation on a set A is a relation from A to A.

Definition 4.5. Given sets $A_1,A_2,\ldots,A_n$ an n-ary relation R is a subset of $A_1\times\ldots\times A_n$.

4.2 Reflexivity, Symmetry, and Transitivity

Definition 4.6. Let R be a relation on a set A.

1. R is **reflexive** if and only if for every $x \in A$, xRx.

2. R is **symmetric** if and only if for every $x, y \in A$, if xRy then yRx.

3. R is **transitive** if and only if for every $x, y, z \in A$, if xRy and yRz then xRz.

Example 4.7. Let $A = \{0, 1, 2, 3\}$ and define the following relations

$$\begin{aligned} R &= \{(0,0), (0,1), (0,3), (1,0), (1,1), (2,2), (3,0), (3,3)\}, \\ S &= \{(0,1), (2,3)\}, \\ T &= \{(0,0)\}. \end{aligned}$$

Are R, S, T transitive, reflexive or symmetric?

R is reflexive and is symmetric, but is not transitive since $(1,0) \in R$ and $(0,3) \in R$, but $(1,3) \notin R$.

S is not reflexive since $(0,0) \notin S$. S is not symmetric since $(0,1) \in S$, but $(1,0) \notin S$. S is transitive since there are no points such that $(x,y) \in S$ and $(y,z) \in S$.

T is transitive, is not reflexive and symmetric.

Example 4.8. Let R be a relation on the real numbers where xRy if and only if $x = y$, then R is symmetric, reflexive, and transitive.

Definition 4.9. Let A be a set and R a relation on A. The **transitive closure** of R is the relation R^t on A that satisfies the following three properties:

1. R^t is transitive.

2. $R \subseteq R^t$.

3. If S is any other transitive relation that contains R, then $R^t \subseteq S$.

Example 4.10. Let $A = \{0, 1, 2, 3\}$ and consider R defined on A as follows:

$$R = \{(0,1), (1,2)\}.$$

Then

$$R^t = \{(0,1), (1,2), (0,2)\}.$$

4.3 Equivalence relations

Definition 4.11. Given a partition of a set A, the **relation induced by the partition**, R, is defined on A as follows: For every $x, y \in A$, $(x, y) \in R$ if x and y are contained in the same subset of the partition.

Example 4.12. Let $A = \{0, 1, 2, 3, 4\}$ and consider the partition

$$\{0, 3, 4\},\ \{1\},\ \{2\}.$$

The relation R induced by this partition is

$$\{(0, 0), (0, 3), (0, 4), (3, 0), (3, 3), (3, 4), (4, 0), (4, 3), (4, 4), (1, 1), (2, 2)\}$$

Theorem 4.13. *Let A be a set with a partition and let R be the relation induced by the partition. Then R is reflexive, symmetric, and transitive.*

Definition 4.14. Let A be a set and R a relation on A, then R is an **equivalence relation** if and only if R is reflexive, symmetric, and transitive.

Note that the above theorem and definition can be combined since they connected with if and only if statements. So we could say that R is an equivalence relation on a set A if and only if R is a relation induced by a partition of A.

Using this if you are asked to show a relation is an equivalence relation one option is to show that there exists a partition of the set such that the relation is induced on it.

Definition 4.15. Suppose A is a set and R is an equivalence relation on A. For each element $a \in A$, the **equivalence class of** a, denoted $[a]$ is the set of all elements $x \in A$ such that $(x, a) \in R$. The element a is called the **representative** for the equivalence class $[a]$.

Another way to think about equivalence classes is with partitions. If we have a set A and a partition, then for some $a \in A$ the equivalence class $[a]$ is the set in the partition where a is in.

Example 4.16. Let $A = \{0, 1, 2, 3, 4\}$ with the partition

$$\{0, 3, 4\},\ \{1\},\ \{2\},$$

then the equivalence classes of A are

$$\begin{aligned}
[0] &= [3] = [4] = \{0, 3, 4\} \\
[1] &= \{1\} \\
[2] &= \{2\}.
\end{aligned}$$

Example 4.17. Consider $\mathbb{Z}$ which can be partitioned into the even and odd numbers, then the equivalence classes of that partition are

$$\begin{aligned}
[0] &= \{\dots, -4, -2, 0, 2, 4, \dots\} \\
[1] &= \{\dots, -3, -1, 1, 3, \dots\}.
\end{aligned}$$

We can also write

$$\mathbb{Z} = [0] \cup [1].$$

Lemma 4.18. *Suppose A is a set, R is an equivalence relation on A, and $a, b \in A$. If $(a, b) \in R$, then $[a] = [b]$.*

Lemma 4.19. *Suppose A is a set, R is an equivalence relation on A, and $a, b \in A$. Either $[a] \cap [b] = \emptyset$ or $[a] \cap [b] = [a]$.*

Definition 4.20. Let $m, n \in \mathbb{Z}$ and let $d \in \mathbb{Z}^+$. We say that m is **congruent** to n modulo d and write

$$m \equiv n \bmod d$$

if and only if

$$d \mid (m - n).$$

Using the definition above we can form partitions of the integers with congruence equivalence classes. We already saw this with the even and odd integers where they are congruent modulo 2.

Example 4.21. Let $d \in \mathbb{Z}^+$ and let R be the relation defined by

$$(x, y) \in R \iff x \equiv y \bmod d,$$

then R is an equivalence relation.

Example 4.22. Let $A = \mathbb{Z} \times (\mathbb{Z} - \{0\})$, then define the relation

$$((a, b), (c, d)) \in R \iff \frac{a}{b} = \frac{c}{d} \iff ad = bc.$$

This relation R is an equivalence class and is a way to define the rational numbers. Consider

$$\left[\frac{1}{2}\right] = \left\{\frac{1}{2}, \frac{2}{4}, \frac{3}{6}, \dots\right\}$$

4.4 Exercises

1. Let $A = \{-1, 1, 2, 4\}$ and $B = \{1, 2\}$ and define relations R and S from A to B as follows: For every $(x, y) \in A \times B$.

$$\begin{aligned} (x, y) \in R &\iff |x| = |y| \\ (x, y) \in S &\iff x - y \text{ is even.} \end{aligned}$$

   State which ordered pairs are in $A \times B$, R, S, $R \cup S$, and $R \cap S$.

2. Let C be the circle relation on the set of real numbers: For every $x, y \in \mathbb{R}$, $(x, y) \in C$ if and only if $x^2 + y^2 = 1$. Determine if C is reflexive, transitive, and symmetric.

3. If R and S are reflexive, then is $R \cap S$ reflexive?

4. If R and S are reflexive, then is $R \cup S$ reflexive?

5. Let $X = \{-1, 0, 1\}$, let $A = \mathcal{P}(X)$, and define R to be a relation on A such that

$$(s,t) \in R \iff \text{the sum of the elements in } s \text{ equals the sum of the elements in } t.$$

   Find the distinct equivalence classes of R.

6. Let $A = \{-4, -3, -2, -1, 0, 1, 2, 3, 4\}$ and define R on A to be

$$(m,n) \in R \iff 4|(m^2 - n^2).$$

   Find the distinct equivalence classes of R.

4.5 Solutions

1. Let $A = \{-1, 1, 2, 4\}$ and $B = \{1, 2\}$ and define relations R and S from A to B as follows: For every $(x,y) \in A \times B$.

$$\begin{aligned} (x,y) \in R &\iff |x|= |y| \\ (x,y) \in S &\iff x - y \text{ is even.} \end{aligned}$$

   State which ordered pairs are in $A \times B$, R, S, $R \cup S$, and $R \cap S$.

$$\begin{aligned} A \times B &= \{(-1,1), (-1,2), (1,1), (1,2), (2,1), (2,2), (4,1), (4,2)\} \\ R &= \{(-1,1), (1,1), (2,2)\} \\ S &= \{(-1,1), (1,1), (2,2), (4,2)\} \\ R \cup S &= \{(-1,1), (1,1), (2,2), (4,2)\} \\ R \cap S &= \{(-1,1), (1,1), (2,2)\}. \end{aligned}$$

2. Let C be the circle relation on the set of real numbers: For every $x, y \in \mathbb{R}$, $(x,y) \in C$ if and only if $x^2 + y^2 = 1$. Determine if C is reflexive, transitive, and symmetric.

   C is not reflexive, consider $(1,1) \notin C$.

   C is symmetric, since x and y are real numbers so we can swap any (x,y) with (y,x).

   C is not transitive. Let $x = 1$, $y = 0$, and $z = 1$, then $(x,y) \in C$ and $(y,z) \in C$, but $(x,z) \notin C$.

3. If R and S are reflexive, then is $R \cap S$ reflexive?

   Yes, since R and S are reflexive, then they both must contain $(x,x) \in A \times A$ for every $x \in A$. So their intersection still contains all those pairs.

4. If R and S are reflexive, then is $R \cup S$ reflexive?

   Yes, similar reasoning to the previous question.

5. Let $X = \{-1, 0, 1\}$, let $A = \mathcal{P}(X)$, and define R to be a relation on A such that

   $$(s, t) \in R \iff \text{the sum of the elements in } s \text{ equals the sum of the elements in } t.$$

   Find the distinct equivalence classes of R.

   First building $\mathcal{P}(X)$ we get

   $$\mathcal{P}(x) = \{\{-1\}, \{0\}, \{1\}, \{-1, 0\}, \{-1, 1\}, \{0, 1\}, \{-1, 0, 1\}, \emptyset\}.$$

   Remember the trick to checking your work on the power set is there should be 2^n elements, so in this case 8 subsets.

   Now to build the equivalence classes we could find all entrys the are related. However for this problem we can assume the relation R is an equivalence relation, so just start building the equivalence classes. So $[\{-1\}]$ is the equivalence class where the sum of the entries is -1 which gives

   $$[\{-1\}] = \{\{-1\}, \{0, -1\}\}.$$

   To get the next equivalence class take an element of $\mathcal{P}(X)$ that is not in $[\{-1\}]$ like $[\{0\}]$ which gives

   $$[\{0\}] = \{\{0\}, \{-1, 1\}, \{-1, 0, 1\}\}.$$

   The next element that is not in $[\{-1\}]$ or $[\{0\}]$ is $\{1\}$, so we can build

   $$[\{1\}] = \{\{1\}, \{0, 1\}\}.$$

   Finally, we are missing $\emptyset$, so we give it its own equivalence class. This gives

   $$\begin{aligned} [\{-1\}] &= \{\{-1\}, \{0, -1\}\} \\ [\{0\}] &= \{\{0\}, \{-1, 1\}, \{-1, 0, 1\}\} \\ [\{1\}] &= \{\{1\}, \{0, 1\}\} \\ [\emptyset] &= \{\emptyset\}. \end{aligned}$$

   as our distinct equivalence classes.

6. Let $A = \{-4, -3, -2, -1, 0, 1, 2, 3, 4\}$ and define R on A to be

   $$(m, n) \in R \iff 4|(m^2 - n^2).$$

   Find the distinct equivalence classes of R.

   Similar to last time we can assume R is an equivalence relation and just start building the classes. So take -4 and we want to find which numbers are related. This gives

   $$[-4] = \{-4, -2, 0, 2, 4\}.$$

   Removing those from A the next element not in $[-4]$ is

   $$[-3] = \{-3, -1, 1, 3\}$$

   These two equivalence classes cover all the elements.

5 Functions

Definition 5.1. A **function** F **from a set** A **to a set** B is a relation with domain A and co-domain B that satisfies the following two properties:

1. For every element $x \in A$, there is an element $y \in B$ such that $(x, y) \in F$.

2. For all elements $x \in A$ and $y \in B$, if $(x, y) \in F$ and $(x, z) \in F$ then $y = z$.

For functions, we frequently use the notation $F(x)$ where $x \in A$ and $F(x) \in B$.

Example 5.2. Define a relation C from $\mathbb{R}$ to $\mathbb{R}$ as follows: For any $(x, y) \in \mathbb{R} \times \mathbb{R}$, $(x, y) \in C$ means that $x^2 + y^2 = 1$.

The domain and co-domain of C are both $\mathbb{R}$.

In its current form C does not satisfy the requirement of being a function since a single input could have two outputs. To see this, consider $x = 0$.

To make C into a function, we could apply a restriction to the co-domain. If we instead defined C from $\mathbb{R}$ to $\mathbb{R}^+$, then C satisfies the requirements of being a function.

Definition 5.3 (Equality of functions)**.** Let $\alpha : A \to B$ and $\beta : A \to B$, then $\alpha = \beta$ if $\alpha(a) = \beta(a)$ for all $a \in A$.

Definition 5.4. Let $f : A \to B$ be a function. We say that the *image* of $a \in A$ under f is the element $b \in B$ such that $f(a) = b$. We can also take the images of sets, for a set $A' \subseteq A$ the image $f(A') = \{f(a) \mid a \in A'\}$.

Definition 5.5. Let $\phi : A \to B$ and $\psi : B \to C$. The composition $\psi\phi$ is the mapping from A to C defined by $(\psi\phi)(a) = \psi(\phi(a))$ for all $a \in A$.

You will likely have seen the notation $(f \circ g)(x) = f(g(x))$ for this class, we will omit the circle and just write $(fg)(x)$.

Note that generally for functions f and g, fg is not the same as gf.

Definition 5.6. A function $\phi : A \to B$ is called *one-to-one* or *injective* if for every $a_1, a_2 \in A$ if $\phi(a_1) = \phi(a_2)$ then $a_1 = a_2$.

A requirement of being a function is that every input has exactly one output, for that function to be one-to-one every output must have exactly one input. Note however that the function ϕ does not need to reach every value of B.

Definition 5.7. A function $\phi : A \to B$ is said to be *onto* B or just onto, if for every $b \in B$ there exists a $a \in A$ such that $\phi(a) = b$. This is also known as *surjective.*

Onto is the property that allows us to say that the function "covers" the set B. If a function is onto, we know that every output has an associated input. However, be careful as being onto might still allow for multiple inputs to have the same output.

Definition 5.8. A function that is both one-to-one and onto is called a bijection.

Bijections form a special type of function since, as you will see formally in the next theorem, they allow for the inverse of the function to be well-defined. One way to think of bijections is that they preserve the "structure" of the set. Or that, both of the two sets are the same under the view of the bijection.

Theorem 5.9. *Given functions $\alpha : A \to B$, $\beta : B \to C$, and $\gamma : C \to D$, then*

1. *$\gamma(\beta\alpha) = (\gamma\beta)\alpha$ (associativity).*
2. *If α and β are one-to-one, then $\beta\alpha$ is one-to-one.*
3. *If α and β are onto, then $\beta\alpha$ is onto.*
4. *If α and β are bijective, then $\beta\alpha$ is bijective.*
5. *If α is a bijection, then there is a function $\alpha^{-1} : B \to A$ such that $(\alpha^{-1}\alpha)(a) = a$ for all $a \in A$ and $(\alpha\alpha^{-1})(b) = b$ for all $b \in B$.*

5.1 Exercises

1. Show that $f : \mathbb{R} \to \mathbb{R}$ defined by $f(x) = x^2$ is neither one-to-one or onto. Give a domain/co-domain restriction that makes it one-to-one and onto.
2. Prove that $f : \mathbb{R} \to \mathbb{R}$ defined by $f(x) = 2x + 1$ forms a bijection.
3. Prove there is a bijection between $\mathbb{N}$ and $\mathbb{Z}$.
4. How many distinct bijections are there between sets $A = \{1, 2, 3\}$ and $B = \{a, b, c\}$? In general, how many bijections are there between two sets of size n?
5. Suppose that α, β, and γ are functions over a set A. If $\alpha\gamma = \beta\gamma$ and γ is a bijection, then prove that $\alpha = \beta$.

5.2 Solutions

1. Show that $f : \mathbb{R} \to \mathbb{R}$ defined by $f(x) = x^2$ is neither one-to-one or onto. Give a domain/co-domain restriction that makes it one-to-one and onto.

   We can do a proof by counter example for both one-to-one and onto.

   Proof. Consider $x = 1$ and $x = -1$, through f both get mapped to 1. Thus, the function f is not one-to-one.

   For onto, consider the output of -1 to reach this we need an x such that $x^2 = -1$ which implies $x = \sqrt{-1}$. We know that $\sqrt{-1}$ is not a real number, so it is not in our domain. Thus f is not onto. □

2. Prove that $f : \mathbb{R} \to \mathbb{R}$ defined by $f(x) = 2x + 1$ forms a bijection.

   Proof. To prove that f forms a bijection we will first show that it is one-to-one, then onto. Let $a_1, a_2 \in \mathbb{R}$ such that $f(a_1) = f(a_2)$ this implies that $2a_1 + 1 = 2a_2 + 1$ which gives $a_1 = a_2$ as desired.

   For showing that it is onto, let $b \in \mathbb{R}$, then we are looking for an $a \in \mathbb{R}$ such that $f(a) = b$. Let $a = (b-1)/2$, then $f(a) = 2(b-1)/2 + 1 = b$. Following f is one-to-one and onto we have that f is a bijection. □

3. Prove there is a bijection between $\mathbb{N}$ and $\mathbb{Z}$.

   Proof. Define $f : \mathbb{N} \to \mathbb{Z}$ such that

$$f(x) = \begin{cases} -(x/2) & \text{if } x \text{ is even,} \\ (x+1)/2 & \text{if } x \text{ is odd.} \end{cases}$$

   The claim to prove is that f is a bijection. First for one-to-one note that if $f(x) \leq 0$, then x is even. If $f(x) > 0$, then x is odd. Now let $a_1, a_2 \in \mathbb{N}$ such that $f(a_1) = f(a_2)$, then we will continue by cases, If $f(a_1) > 0$, then a_1 and a_2 are both odd giving that

$$f(a_1) = f(a_2) \implies \frac{a_1 + 1}{2} + \frac{a_2 + 1}{2} \implies a_1 = a_2.$$

   If $f(a_2) \leq 0$, then a_1 and a_2 are both even giving that

$$f(a_1) = f(a_2) \implies -(a_1/2) = -(a_2/2) \implies a_1 = a_2.$$

   Thus f is one-to-one.

   For onto let $b \in \mathbb{Z}$ if $b > 0$, then consider $2b - 1$ which is an odd natural number. Now $f(2b-1) = (2b - 1 + 1)/2 = b$. If $b \leq 0$, then consider $-2b$ which is an even natural number. With this we have, $f(-2b) = -(-2b/2) = b$. This gives that f is also onto. Therefore, f is a bijection. □

4. How many distinct bijections are there between sets $A = \{1, 2, 3\}$ and $B = \{a, b, c\}$? In general, how many bijections are there between two sets of size n?

   Between sets A and B there are 6 possible bijections. To see this, first consider where we can map 1 from set A. It can go to either a, b, or c giving 3 possible choices. Now 2 can be mapped to either of the two choices that 1 was not mapped to, giving an additional two choices. Finally, 3 is fully locked in to the final choice. This give $3(2)(1) = 6$ total bijections.

   The same argument as above can be used to generalize the problem. In this case the first choice will have n options, the ith choice will have $n - i$ options and the final choice will have 1 option. This gives a total of $n!$ different bijections.

5. Suppose that α, β, and γ are functions over a set A. If $\alpha\gamma = \beta\gamma$ and γ is a bijection, then prove that $\alpha = \beta$.

   To show that two functions are equal, we need to show that for every $x \in A$ we have that $\alpha(x) = \beta(x)$. In this case, a contradiction proof works fairly well.

   Proof. Assume for contradiction that $\alpha \neq \beta$, then there exists a $x \in A$ such that $\alpha(x) \neq \beta(x)$. Following γ is a bijection, there exists a $y \in A$ such that $\gamma(y) = x$. From assumptions, we have that, $\alpha(\gamma(y)) = \beta(\gamma(y))$ which implies that $\alpha(x) = \beta(x)$. Thus, we have a contradiction, since $\alpha(x)$ can't be both equal and not equal to $\beta(x)$. $\square$